

Jomo Kenyatta University of Agriculture and Technology
ECE / EEE Departments
EEE 2504 Quantum Electronics Laboratory Exercise
Familiarisation with Photonic Devices

Exercise 1: Optical Sources

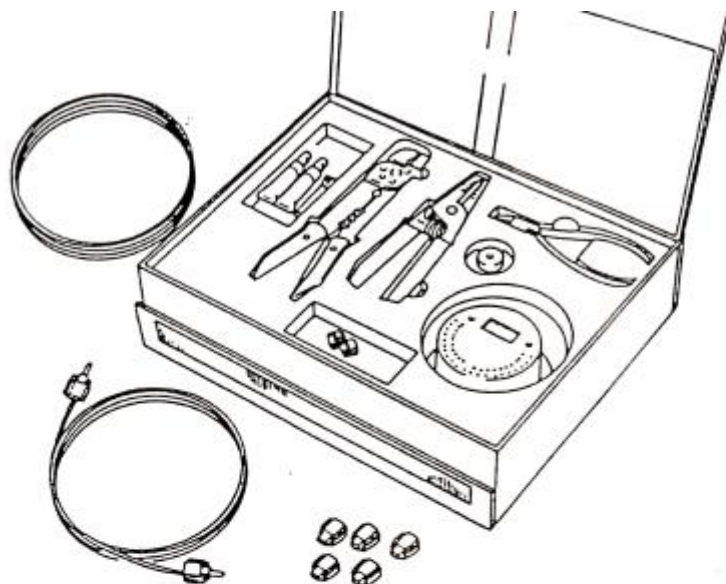
1. The optical laboratory kit contains a number of optical sources and associated equipment. Observe the equipment noting the light outlets and types of connectors used.
2. **Deliverables:**
 - a. Prepare a table showing the wavelength(s) of each device. Evaluate the associated frequency and indicate the position in the electromagnetic spectrum.
 - b. Indicate the precautions to be taken when working with optical sources.

ITEM	MODULE No.	WAVELENGTH(S) (nm)	FREQUENCY (THz)	BAND
1	LDP-10			
2	ME-10-13			
3	ME-10-85			
4	MQ-10-02			
5				

Add rows as required.

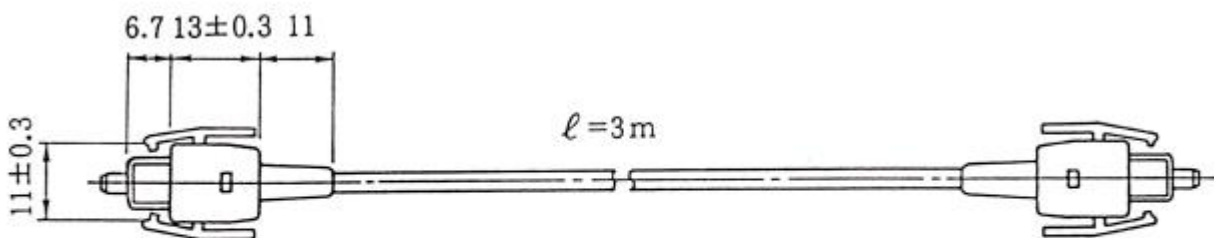
Exercise 2: Optical Fibre and Fibre Assembly Kit

1. The optical laboratory kit contains some optical fibres, some with termination and some without termination and some connectors.
2. From the un-terminated fibre, cut-off a piece approximately 300mm long:
 - a. Clean and polish the cut end using the tools in the assembly kit.
 - b. Using a torch as a source of light, try to pass light from one end of the piece of fibre to the other.
 - c. Estimate the external diameter of the fibre, including the coating.
 - d. Strip the coating from one end of the fibre and estimate the diameter of the core and cladding.
3. Observe the fibre termination kit and the optical connector parts.
4. On the piece of fibre cut-off in (2) above, strip one end and go through the process of assembling into the connector but without applying adhesive or crimping the ferrule.
5. **Deliverables:**
 - a. List the contents of the assembly kit and indicate the purpose of each.
 - b. List the components of the fibre connector and indicate the purpose of each.
 - c. Indicate how transmission losses can arise from optic fibre termination, optical fibre splicing and other installation activities.
 - d. From the specifications given for the optical fibre, would you expect it to be single mode or multi-mode?
 - e. Evaluate the numerical aperture and acceptance angle for the optical fibre.
 - f. If the input power to the cable is 2mW at a wavelength of 660nm, what will be the output power after transmission in the cable for 400m?



① Optical fiber cord with optical connector CF1000	1 cord
② Optical connector parts	10 pcs.
③ Optical fiber cord CEE-AP-100/02	10 m
④ Sandpaper	1 pc.
⑤ Calking tool	1 pc.
⑥ Polishing jig	1 pc.
⑦ Hot plate heating type end face processing device	1 unit
⑧ Wire stripper	1 pc.
⑨ Nipper	1 pc.
10 Adhesive	1 set
11 FC connector adapter	2 pcs.

Figure 1: Fibre Termination Kit



- (1) Optical connector CF-1000
- (2) Optical fiber cord CEE-AP-100/02

Figure 2: Fibre with Connectors Assembled

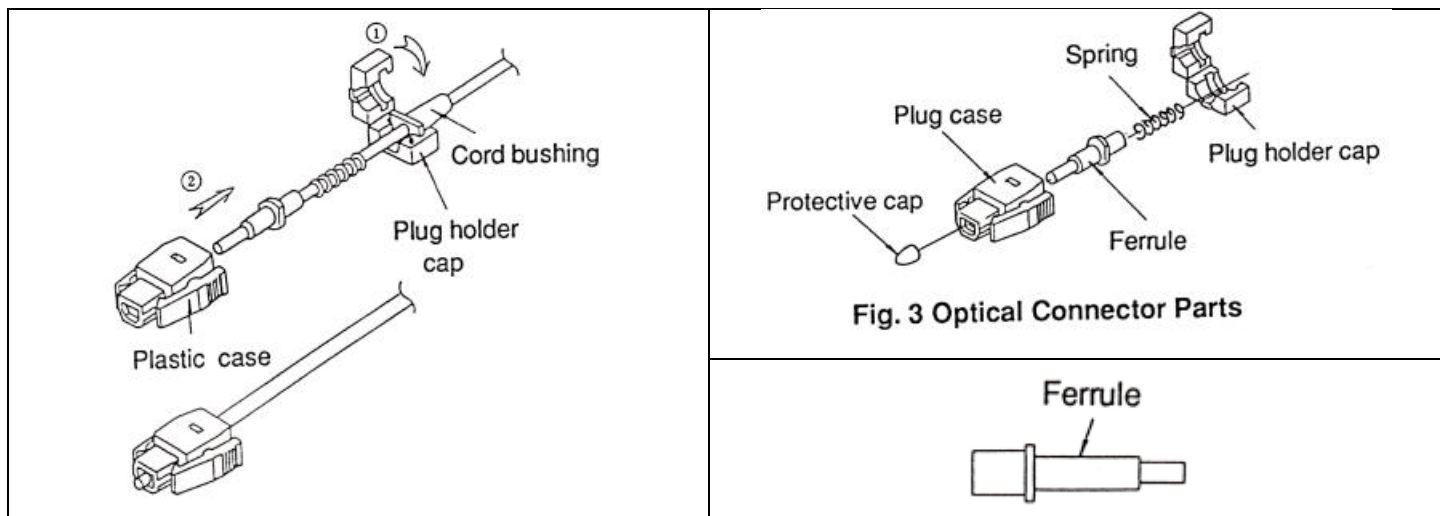


Figure 3: Fibre Connectors Blown up View

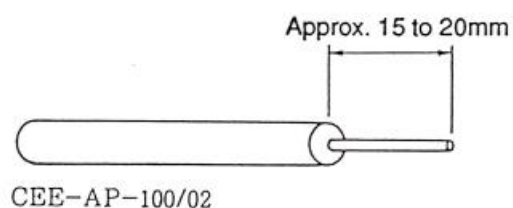


Fig. 10 Optical Fiber with Sheath Removed

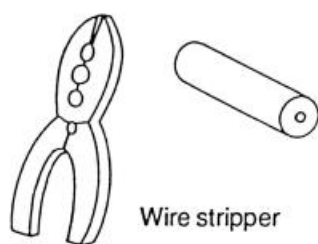


Fig. 11 Spring Inserting

Figure 4: Optical Fibre Stripping and Connection to Ferrule

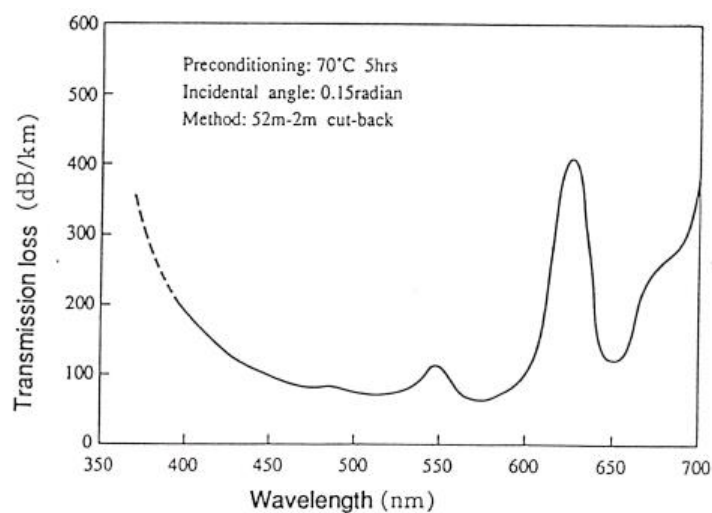


Fig. 5 Wavelength Characteristic of Optical Fiber Cord for CF-1000

Figure 5: Optical Fibre Transmission Loss

CEE-AP-100/02 has the structure shown in Fig. 4 and the core and clad are made of plastics.

It is a step index type optical fiber with refractive indexes of 1.495 for the core and 1.402 for the clad. OPT-210 has such features as (1) N.A.=0.47 and core diameter is large at 0.98mm and much light can be easily connected, but (2) transmission loss is large due to plastics, and (3) band is narrow. For this reason, it is best suited for short-distance, small-capacity optical fiber for practical training.

Fig. 5 shows the wavelength characteristic of transmission loss of the optical fiber cord. The light emission wavelength of the E/O transducer OPT-200 is 670nm, and efficient signal transmission can be made when combined with plastic optical fiber OPT-210.

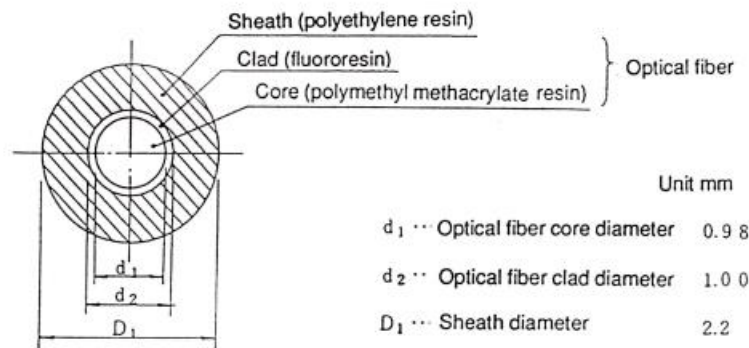


Fig. 4 Structure of Optical Fiber Cord for CF-1000

Figure 6: Optical Fibre Specifications